

Sparse Expert Routing on MobileNetV3-Small for Tomato Leaf Disease Classification

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ABSTRACT Automatic tomato leaf disease classification in field images is limited by visually similar disease categories, non-uniform background conditions, and the active computational cost of edge-side inference. MobileNetV3-Small provides a compact feature extractor for image analysis of plant leaf, but its single dense classifier head has limited capacity to separate disease categories. This paper introduces MobileNetV3-Small-MoE, which retains MobileNetV3-Small as the shared backbone and adds a sparse mixture-of-experts classifier head. For each image, the gate activates only the top- k experts, increasing classifier-side capacity without activating the full expert pool. On a tomato subset derived from PlantDoc with 2843 images and eight categories, the proposed model improves accuracy from 0.8140 to 0.8596 and Macro-F1 from 0.7799 to 0.8275 over MobileNetV3-Small. The reported sparse configuration uses four experts with top-2 routing. Linear gating achieves performance close to MLP gating with a simpler routing function. The parameter count increases from 1.526M to 3.484M, while FLOPs increase from 0.0555G to 0.0634G because two experts are activated per image. These results show that sparse classifier-side routing provides a controlled capacity cost trade-off for tomato leaf disease classification.

INDEX TERMS Tomato leaf disease classification, MobileNetV3-Small, mixture-of-experts, sparse routing, top- k gating, edge image classification.

I. INTRODUCTION

AUTOMATIC tomato leaf disease classification is an image-based diagnosis task in smart agriculture. The input leaf images are affected by non-uniform illumination, background texture interference, scale variation, and visually similar disease categories. In a camera-side or edge processing pipeline, the classifier must preserve useful leaf-image features for disease-category separation while maintaining low active computational cost on the local device. This creates a trade-off between classifier capacity and active computational cost [1].

Deep convolutional networks have shown strong performance for leaf disease classification when sufficient labeled images are available [2]–[4]. However, field images are more challenging than controlled leaf-image datasets because disease-related leaf regions are affected by background texture interference, non-uniform illumination, and partial leaf occlusion. PlantDoc was introduced to support plant disease classification under more realistic visual conditions, including images with non-uniform background conditions and natural acquisition variability [5]. For tomato leaf disease

classification, local patterns such as spots, blight regions, mosaic texture, and yellowing may produce similar color and texture responses across disease categories.

MobileNetV3-Small is a suitable backbone when the local device has a constrained processing budget, following the broader line of compact CNNs such as MobileNetV2, MobileNetV3, EfficientNet, ShuffleNetV2, GhostNet, and SqueezeNet [6]–[12]. MobileNetV3 was designed for mobile inference using hardware-aware architecture search, NetAdapt, and compact convolutional blocks [7]. Nevertheless, a compact backbone with a single dense classifier head may not provide sufficient capacity for disease-category separation when local leaf-image patterns are similar. Replacing the full backbone with a heavier network would increase costs across the entire feature-extraction path. Therefore, this work retains MobileNetV3-Small as the shared feature extractor and places sparse conditional capacity only in the classifier head.

This paper studies a sparse mixture-of-experts (MoE) classification head on top of MobileNetV3-Small, following the conditional-computation principle introduced by modular ex-

pert models and sparse MoE layers [13]–[15]. For each input image, the backbone produces a shared leaf-image feature vector, and a routing function selects only k experts from a pool of E expert heads. The stored classifier capacity is increased through multiple expert heads, while the active computational cost is controlled by the number of selected experts.

The proposed model is evaluated on a tomato subset derived from PlantDoc [5]. The experimental results show that MobileNetV3-Small-MoE improves both accuracy and Macro-F1 compared with the MobileNetV3-Small baseline. The ablation over the number of experts and the top- k routing width shows that $E = 4$ and $k = 2$ provide the best tested configuration. The comparison between Linear Gating and MLP Gating shows that a linear router is sufficient under the present feature space and training protocol. The complexity analysis shows the main trade-off: the parameter count increases because all expert heads must be stored, whereas FLOPs increase only slightly because only a subset of experts is activated per image.

The contributions of this paper are summarized as follows. First, the paper studies tomato leaf disease classification in an edge-processing scenario, where leaf-image features for disease-category separation and the active computational cost are jointly considered. Second, it develops a MobileNetV3-Small-MoE classifier in which MobileNetV3-Small serves as the shared feature extractor, with sparse expert routing applied only to the classifier head. Third, it analyzes the routing mechanism through ablations on linear and MLP gating, expert pool size, and top- k activation width. Fourth, we evaluate classification performance and complexity using the same protocol, showing that the proposed model improves Macro-F1 by increasing the stored classifier capacity while keeping the active computational cost close to that of the MobileNetV3-Small baseline.

II. RELATED WORK

A. IMAGE-BASED PLANT DISEASE CLASSIFICATION

Image-based plant disease classification is a camera diagnosis task in agricultural monitoring. CNN-based classifiers have shown strong performance on controlled leaf-image datasets, where background and illumination are relatively stable [2]. Transfer learning further improves classification when labeled agricultural images are limited, because pre-trained CNN and residual representations can be adapted to plant leaf image analysis [3], [4], [16]. Recent surveys show that leaf disease classification must handle data variability, limited annotation, and deployment constraints, rather than only accuracy on controlled datasets [1], [17].

Field-image datasets make leaf disease classification closer to deployment. PlantDoc provides plant disease images under natural acquisition conditions, where disease-related leaf regions must be separated from non-uniform background conditions and background texture interference [5]. Recent lightweight, attention-based, and hybrid tomato disease classifiers improve local feature extraction under small inter-

category differences and large intra-category variation [18], [19]. These studies strengthen feature extraction, but they do not explicitly isolate classifier-side conditional computation under active computational cost constraints.

B. LIGHTWEIGHT CNNs FOR EDGE PROCESSING

Lightweight CNNs are important for edge processing in smart agriculture because inference may run on mobile devices, embedded boards, or local camera-side units. MobileNetV2 reduces computation through inverted residuals and linear bottlenecks [6]. MobileNetV3 targets mobile CPU inference through hardware-aware architecture search and NetAdapt [7], and MobileNetV3-Small has also been used for edge-oriented plant disease detection [8]. EfficientNet balances depth, width, and resolution through compound scaling [9]. Other compact backbones address deployment cost from different design points: ShuffleNetV2 emphasizes implementation-aware CNN design beyond FLOPs, GhostNet generates additional feature maps through small operations, and SqueezeNet reduces parameter storage using fire modules [10]–[12].

For plant disease classification, lightweight CNNs have been evaluated on embedded hardware, in compact agricultural classifiers, and in edge-device disease-detection scenarios [20]–[22]. These methods mainly reduce the cost of the backbone or convolutional modules. In contrast, the proposed work keeps MobileNetV3-Small as the shared feature extractor and introduces additional capacity only at the classifier side. This distinction is important because the proposed model targets low active computational cost, not a smaller storage footprint.

C. SPARSE MIXTURE-OF-EXPERTS AND CONDITIONAL COMPUTATION

Mixture-of-experts models use multiple parameter branches and a routing function to assign each input to a subset of experts [13], [14]. Sparse MoE extends this principle by activating only part of the expert pool per input, thereby increasing stored capacity without proportional growth in active computational cost [15]. Large sparse models show that routing can decouple parameter capacity from active computational cost in language and vision transformer systems [23]–[25]. Recent vision MoE analysis further shows that routing design affects both classification performance and computation allocation [26].

These MoE systems are mainly designed for token-level routing in large transformer models. Their dataset scale, routing granularity, and load-balancing assumptions do not align with compact CNN-based plant disease classification. Our work uses MoE in a bounded form; a small expert pool is attached only to the classifier head of MobileNetV3-Small, and top- k routing controls how many expert heads are active for each image.

The above related work leaves a specific gap. Plant disease classifiers improve leaf disease classification under non-uniform image conditions, but most do not isolate the compu-

tational cost at the classifier level. Lightweight CNNs reduce feature-extraction cost, but their final classifier is usually dense and input-independent. Sparse MoE provides conditional computation, but its dominant use cases are large transformer systems rather than compact CNN classifiers. The proposed MobileNetV3-Small-MoE is positioned at this intersection. The backbone is fixed to preserve the low-cost feature extraction path. Conditional computation is introduced only through sparse expert routing at the classifier head. Our experimental results show that the model increases the stored classifier capacity and keeps the active computational cost close to that of the MobileNetV3-Small baseline by activating only a subset of experts per image.

III. PROPOSED MOBILENETV3-SMALL-MOE MODEL

A. PROBLEM FORMULATION

Let $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the training set, where $\mathbf{x}_i$ is an RGB tomato leaf image and $y_i \in \{1, \dots, C\}$ is the category label. In this work, $C = 8$, corresponding to seven disease categories and one healthy state. The objective is to learn a classifier $F_{\Theta}(\cdot)$ that predicts the posterior probability vector $\hat{\mathbf{p}}_i \in \mathbb{R}^C$ while maintaining a controlled active computational cost.

Fig. 1 summarizes the proposed architecture and separates the shared backbone from the classifier-side conditional computation.

The baseline MobileNetV3-Small classifier can be written as

$$\hat{\mathbf{p}}_i^{\text{base}} = \text{softmax}(\mathbf{W}_c f_{\theta}(\mathbf{x}_i) + \mathbf{b}_c), \quad (1)$$

where $f_{\theta}(\cdot)$ denotes the MobileNetV3-Small feature extractor. Equation (1) uses a single dense classifier path for all samples; the same classifier parameters are applied regardless of leaf-image condition or background condition.

The proposed model keeps the shared feature extractor and replaces the single dense classifier head with a sparse expert head. The shared feature vector is

$$\mathbf{u}_i = f_{\theta}(\mathbf{x}_i), \quad \mathbf{u}_i \in \mathbb{R}^d. \quad (2)$$

Equation (2) defines the common encoder output used by the gate and all expert heads; therefore, the conditional computation is introduced after feature extraction rather than by duplicating the full backbone.

B. EXPERT HEAD ARCHITECTURE

The proposed MoE module operates at the classifier side. Each expert is implemented as a lightweight feature-transformation head. This architecture keeps the MobileNetV3-Small feature extractor shared across all input images and confines conditional computation to the final representation stage.

For the general case, expert e is written as a two-layer fully connected head:

$$\mathbf{z}_{i,e} = \varphi_e(\text{Norm}_e(\mathbf{A}_e \mathbf{u}_i + \mathbf{a}_e)), \quad \mathbf{z}_{i,e} \in \mathbb{R}^{r_e}, \quad (3)$$

where $\mathbf{A}_e \in \mathbb{R}^{r_e \times d}$, $\mathbf{a}_e \in \mathbb{R}^{r_e}$, r_e is the hidden width of expert e , $\varphi_e(\cdot)$ is the activation function, and $\text{Norm}_e(\cdot)$ denotes the optional normalization layer. Equation (3) defines the internal transformation performed by one expert after the shared MobileNetV3-Small feature vector is obtained; the expert does not reprocess the image but only reshapes the classifier-side representation.

The expert output is then

$$q_e(\mathbf{u}_i) = \mathbf{B}_e \text{Drop}_{p_e}(\mathbf{z}_{i,e}) + \mathbf{b}_e, \quad q_e(\mathbf{u}_i) \in \mathbb{R}^{d_o}, \quad (4)$$

where $\mathbf{B}_e \in \mathbb{R}^{d_o \times r_e}$, $\mathbf{b}_e \in \mathbb{R}^{d_o}$, p_e is the dropout rate, and d_o is the expert output dimension. Equation (4) maps the hidden expert representation to a common output space so that the selected experts can be aggregated by the sparse routing coefficients.

If the implemented expert is a single linear projection, Eq. (3) can be omitted, and the expert becomes

$$q_e(\mathbf{u}_i) = \mathbf{B}_e \mathbf{u}_i + \mathbf{b}_e, \quad q_e(\mathbf{u}_i) \in \mathbb{R}^{d_o}. \quad (5)$$

Equation (5) corresponds to the lowest-complexity expert head; in this case, the gain of MoE comes only from routing among multiple linear classifier-side projections.

In the current implementation, the exact expert configuration is as follows: each expert is a two-layer MLP consisting of a Linear layer $d \rightarrow r_e$, followed by LayerNorm, GELU activation, Dropout with rate $p_e = 0.1$, and a final Linear layer $r_e \rightarrow d_o$. The MobileNetV3-Small feature dimension is $d = 576$ in the current backbone. The expert hidden width is $r_e = 1024$, shared across all experts. The expert output dimension is $d_o = 576$. GELU is used as the activation function. Layer Normalization is applied after the first linear transformation. Dropout with rate $p_e = 0.1$ is used for regularization.

C. SPARSE EXPERT ROUTING

The MoE classifier contains E expert heads $\{q_e(\cdot)\}_{e=1}^E$, each following the expert realization in Eqs. (3)–(4), or the linear form in Eq. (5) when a single-layer expert is used. The role of the gate is not to classify the image directly but to select which expert heads should process the shared feature vector $\mathbf{u}_i$.

A gating function computes the routing score vector and routing probabilities as

$$\mathbf{s}_i = g_{\phi}(\mathbf{u}_i), \quad \boldsymbol{\pi}_i = \text{softmax}(\mathbf{s}_i), \quad (6)$$

where $\boldsymbol{\pi}_i = [\pi_{i,1}, \dots, \pi_{i,E}]^T$. Equation (6) converts the backbone feature into expert-selection weights, allowing different samples to use different classifier-side branches.

For each image, only the k experts with the largest routing probabilities are activated:

$$\mathcal{K}_i = \text{TopK}(\boldsymbol{\pi}_i, k). \quad (7)$$

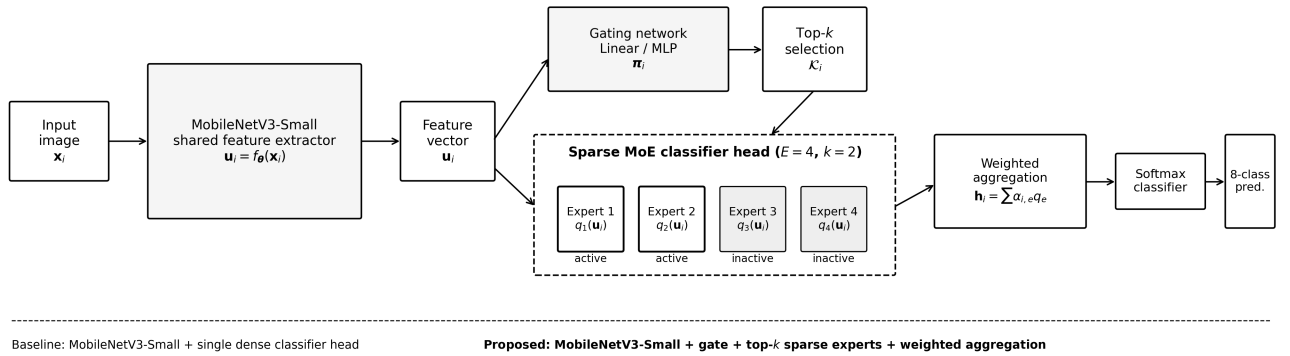
Equation (7) enforces sparse activation; the number of active expert branches is controlled by k instead of the total expert pool size E .

TABLE 1. Positioning of the proposed model against representative technical routes.

Technical route	Representative works	Main improvement	Remaining limitation addressed here
Plant disease classification	[2]–[5], [18], [19], [22], [27]	Improves leaf disease classification under controlled or field image conditions.	Classifier-side active computational cost is not isolated as a design variable.
Lightweight CNN backbone	[6], [8]–[12]	Reduces feature-extraction cost for mobile or embedded inference.	The classifier head usually remains dense and input-independent.
Sparse expert routing	[14], [15], [23]–[26]	Decouples stored capacity from active computational cost through routing.	Existing designs mainly target large transformer systems.
Proposed design	This work	Adds sparse classifier-side expert routing to MobileNetV3-Small.	Evaluates a controlled capacity-cost trade-off for tomato leaf disease classification.

Overall architecture of the proposed MobileNetV3-Small-MoE classifier

The backbone is shared for all samples; conditional computation is introduced only in the classifier head.

**FIGURE 1.** Overall architecture of the proposed MobileNetV3-Small-MoE classifier. The MobileNetV3-Small backbone extracts a shared feature vector for all images. The proposed control point is the classifier head, where a linear or MLP gate produces routing probabilities, top- k selection activates two experts from a four-expert pool, and the selected expert outputs are aggregated before the softmax classifier.

The sparse routing coefficient of expert e is defined as

$$\alpha_{i,e} = \frac{\pi_{i,e} \mathbb{I}(e \in \mathcal{K}_i)}{\sum_{j \in \mathcal{K}_i} \pi_{i,j}}, \quad e = 1, \dots, E. \quad (8)$$

Equation (8) removes inactive experts from the forward path and renormalizes the selected routing weights, so the aggregated representation remains comparable across samples.

The MoE representation and final class posterior are

$$\mathbf{h}_i = \sum_{e=1}^E \alpha_{i,e} q_e(\mathbf{u}_i), \quad \hat{\mathbf{p}}_i = \text{softmax}(\mathbf{W}_o \mathbf{h}_i + \mathbf{b}_o). \quad (9)$$

Equation (9) combines only the selected expert outputs and produces the class probability vector; this is the mechanism that increases stored classifier capacity while preserving a sparse active path.

D. LINEAR AND MLP GATING

Two gating structures are evaluated. The first one is a linear gate:

$$g_{\phi}^{\text{lin}}(\mathbf{u}_i) = \mathbf{W}_g \mathbf{u}_i + \mathbf{b}_g. \quad (10)$$

Equation (10) gives the lowest-complexity routing function and is suitable when the MobileNetV3-Small feature space is already sufficiently structured for expert assignment.

The second one is an MLP gate:

$$g_{\phi}^{\text{mlp}}(\mathbf{u}_i) = \mathbf{W}_2 \sigma(\mathbf{W}_1 \mathbf{u}_i + \mathbf{b}_1) + \mathbf{b}_2, \quad (11)$$

where $\sigma(\cdot)$ is the nonlinear activation function. Equation (11) increases routing flexibility, but it also adds parameters and operations before the top- k selection.

E. TRAINING OBJECTIVE AND EVALUATION METRICS

The load-bearing training objective is the cross-entropy loss:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N} \sum_{i=1}^N \log \hat{p}_{i,y_i}. \quad (12)$$

Equation (12) directly penalizes incorrect tomato disease classification and provides the main gradient signal for the classifier head.

To stabilize routing and reduce expert underutilization, the auxiliary load-balancing term is included in the training loss as

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}}, \quad (13)$$

where λ_{bal} controls the strength of the routing-balance penalty. Equation (13) keeps the classification objective dominant while discouraging collapse of the sparse router to a small subset of experts.

The classification accuracy is computed as

$$\text{Acc} = \frac{1}{N_{\text{te}}} \sum_{i=1}^{N_{\text{te}}} \mathbb{I} \left(\arg \max_c \hat{p}_{i,c} = y_i \right). \quad (14)$$

Equation (14) measures the overall decision correctness over the test set.

Because the task contains multiple disease classes and may contain uneven class frequencies, Macro-F1 is also reported [28], [29]:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \frac{2P_c R_c}{P_c + R_c + \epsilon}, \quad (15)$$

where P_c and R_c are the precision and recall of class c , respectively, and ϵ avoids numerical division by zero. Equation (15) gives equal weight to each class, which makes it more informative than accuracy when the classifier may perform unevenly across tomato disease categories.

To inspect expert utilization, the empirical activation ratio of expert e is computed as

$$\rho_e = \frac{1}{N_{\text{tr}}} \sum_{i=1}^{N_{\text{tr}}} \mathbb{I}(e \in \mathcal{K}_i), \quad e = 1, \dots, E. \quad (16)$$

Equation (16) measures how often each expert is selected during training; balanced usage indicates that the router does not collapse to a single expert, but it does not by itself prove disease-category-specific expert behavior.

F. COMPLEXITY MODEL

The stored parameter count of the MoE model can be decomposed as

$$\mathcal{P}_{\text{MoE}} = \mathcal{P}_{\text{bb}} + \mathcal{P}_{\text{gate}} + \sum_{e=1}^E \mathcal{P}_e + \mathcal{P}_{\text{out}}, \quad (17)$$

where $\mathcal{P}_{\text{bb}}$, $\mathcal{P}_{\text{gate}}$, $\mathcal{P}_e$, and $\mathcal{P}_{\text{out}}$ denote the parameters of the backbone, gate, expert e , and output classifier, respectively. Equation (17) explains why the proposed model increases memory footprint: all expert parameters must be stored even though only a subset is activated per image.

The active computational cost per image is approximated by

$$\mathcal{F}_{\text{MoE}}(\mathbf{x}_i) = \mathcal{F}_{\text{bb}} + \mathcal{F}_{\text{gate}} + \sum_{e \in \mathcal{K}_i} \mathcal{F}_e + \mathcal{F}_{\text{out}}. \quad (18)$$

Equation (18) gives the computational reason for top- k routing: the active computational cost depends on the selected experts, not on the full expert pool.

For a two-layer expert with hidden width r_e and output dimension d_o , the number of parameters of expert e is

$$\mathcal{P}_e = r_e d + r_e + d_o r_e + d_o + \mathcal{P}_{\text{norm},e}, \quad (19)$$

where $\mathcal{P}_{\text{norm},e}$ is zero if no normalization is used and equals the number of trainable scale and shift parameters when a normalization layer is used. Equation (19) shows that the expert pool increases stored memory linearly with the number of experts.

The active expert-side FLOPs for image $\mathbf{x}_i$ can be approximated as

$$\mathcal{F}_{\text{expert}}(\mathbf{x}_i) = \sum_{e \in \mathcal{K}_i} (2r_e d + 2d_o r_e), \quad (20)$$

where multiplication and addition are counted separately. Equation (20) makes explicit that the active expert computation depends on $k = |\mathcal{K}_i|$, not on the full expert pool size E .

For Linear Gating, the routing parameter count is

$$\mathcal{P}_{\text{gate}}^{\text{lin}} = E d + E. \quad (21)$$

Equation (21) explains why Linear Gating adds only a small routing overhead when the number of experts is small.

For MLP Gating with hidden width r_g , the routing parameter count is

$$\mathcal{P}_{\text{gate}}^{\text{mlp}} = r_g d + r_g + E r_g + E. \quad (22)$$

Equation (22) shows that the MLP gate increases routing flexibility at the cost of additional classifier-side parameters and operations. Therefore, the comparison between Linear Gating and MLP Gating should be interpreted jointly with classification performance and routing overhead.

IV. EXPERIMENTAL RESULTS

A. DATASET AND EXPERIMENTAL SETUP

The experiments use a tomato subset derived from PlantDoc. The subset contains 2843 tomato leaf images with non-uniform background conditions and background texture interference. The classification task contains eight categories, including seven tomato disease categories and one healthy state. The reported results are averaged over five independent runs and are presented as mean $\pm$ standard deviation.

The dataset was partitioned into training, validation, and test subsets following an 80:10:10 ratio. Input RGB images were resized to 224×224 . During training, data augmentation included random cropping, horizontal flipping, vertical flipping, random rotation, and shift-scale-rotate transformations. All models were optimized using the AdamW optimizer [30] with an initial learning rate of 1×10^{-3} , weight decay of 1×10^{-3} , and a batch size of 32 for 300 epochs. To ensure reproducibility, all experiments were repeated using five random seeds: 42, 43, 44, 45, and 46. All experiments were conducted on an NVIDIA GB10 GPU, and FLOPs were measured using the THOP library [31]. These settings ensure reproducibility and provide a consistent basis for interpreting the reported mean and standard deviation values.

B. COMPARISON WITH MOBILENETV3-SMALL BASELINE

Table 2 compares the baseline MobileNetV3-Small with the proposed MobileNetV3-Small-MoE on the tomato disease

TABLE 2. Classification performance of MobileNetV3-Small and MobileNetV3-Small-MoE.

Model	Accuracy	Macro-F1
MobileNetV3-Small	0.8140 ± 0.0250	0.7799 ± 0.0276
MobileNetV3-Small-MoE	0.8596 ± 0.0124	0.8275 ± 0.0153

TABLE 3. Effect of gating structure on classification performance.

Gating type	Accuracy	Macro-F1
Linear	0.8596 ± 0.0124	0.8275 ± 0.0153
MLP	0.8596 ± 0.0074	0.8300 ± 0.0082

classification task. The proposed model improves the mean accuracy from 0.8140 ± 0.0250 to 0.8596 ± 0.0124 and the mean Macro-F1 score from 0.7799 ± 0.0276 to 0.8275 ± 0.0153 . The lower standard deviation values indicate more stable cross-run behavior under the present training protocol.

The larger improvement in Macro-F1 than in accuracy suggests that the sparse expert mechanism improves the balance of predictions across disease categories. This result indicates that the MoE classifier head improves class discrimination over the dense baseline classifier under the tested dataset and training protocol. Further cross-dataset validation is still required before broader generalization can be claimed.

C. EFFECT OF GATING STRUCTURE

Table 3 presents the impact of different gating structures on the proposed model's classification performance. Both Linear Gating and MLP Gating achieve the same mean accuracy of 0.8596, indicating that introducing a nonlinear gating mechanism does not yield a measurable improvement in overall classification accuracy in the current experimental setting. Although MLP Gating attains a slightly higher mean Macro-F1 score (0.8300) compared with Linear Gating (0.8275), the observed difference remains relatively small with respect to the reported standard deviations. Consequently, the results do not provide sufficient evidence to conclude that either gating strategy offers a statistically significant advantage over the other.

These findings suggest that nonlinear routing does not result in a clear performance separation within the proposed MobileNetV3-Small-MoE framework. Compared with MLP Gating, Linear Gating employs a simpler routing mechanism with fewer computational operations before the top- k expert selection stage. A plausible explanation is that the feature representations generated by the MobileNetV3-Small backbone are sufficiently discriminative, enabling a linear routing function to separate expert assignments effectively. Therefore, considering both routing efficiency and implementation simplicity, Linear Gating can be regarded as the more practical and computationally efficient choice for the current architecture.

TABLE 4. Ablation study on the number of experts and top- k routing.

Number of experts E	Activated experts k	Accuracy	Macro-F1
2	1	0.8428 ± 0.0237	0.8063 ± 0.0265
2	2	0.8540 ± 0.0213	0.8255 ± 0.0219
3	1	0.8393 ± 0.0196	0.8039 ± 0.0201
3	2	0.8330 ± 0.0330	0.7970 ± 0.0391
4	1	0.8386 ± 0.0259	0.8106 ± 0.0303
4	2	0.8596 ± 0.0074	0.8290 ± 0.0082
5	1	0.8414 ± 0.0112	0.8084 ± 0.0148
5	2	0.8302 ± 0.0358	0.7925 ± 0.0438
6	1	0.8407 ± 0.0168	0.8078 ± 0.0183
6	2	0.8421 ± 0.0277	0.8087 ± 0.0351
6	3	0.8470 ± 0.0296	0.8114 ± 0.0314
7	1	0.8463 ± 0.0130	0.8115 ± 0.0102
7	2	0.8372 ± 0.0432	0.8071 ± 0.0462
7	3	0.8421 ± 0.0134	0.8091 ± 0.0150
8	1	0.8393 ± 0.0132	0.8030 ± 0.0148
8	2	0.8456 ± 0.0055	0.8087 ± 0.0081
8	3	0.8239 ± 0.0426	0.7885 ± 0.0498
8	4	0.8414 ± 0.0279	0.8108 ± 0.0314

D. EFFECT OF NUMBER OF EXPERTS AND TOP-K

Table 4 reports the ablation over the number of experts and the number of activated experts. The best-performing configuration is $E = 4$ and $k = 2$, achieving 0.8596 accuracy and 0.8290 Macro-F1. Increasing the number of experts beyond four does not produce a consistent gain.

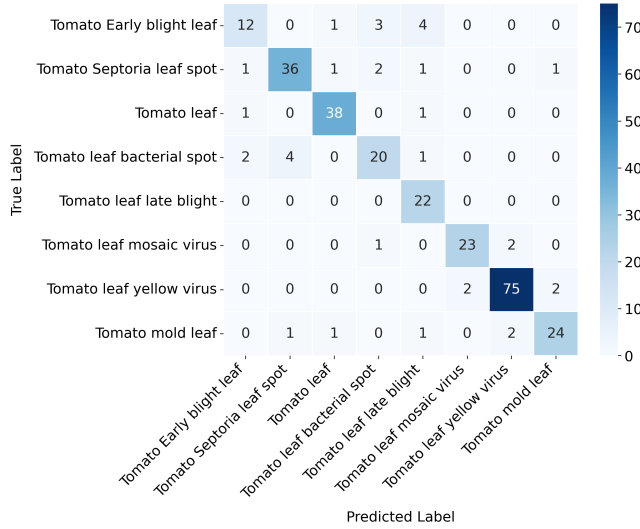
To visualize the ablation trend in Table 4, Fig. 3 shows the Macro-F1 heatmap over the expert number E and the top- k routing width. This figure provides a direct view of the operating point of the sparse MoE design and complements the numerical comparison in Table 4.

The observation is that $E = 4$ and $k = 2$ yield the best measured operating point, as shown in Fig. 3. Compared with $E = 4, k = 1$, activating two experts increases Macro-F1 from 0.8106 to 0.8290. This result indicates that a single active expert is insufficient for some images with mixed leaf-image patterns and background-texture interference. Compared with larger expert pools, the four-expert configuration is also more stable, as reflected by the lower standard deviation. The implication is that the expert pool should be selected based on data scale and active computational cost, rather than by increasing mechanically E .

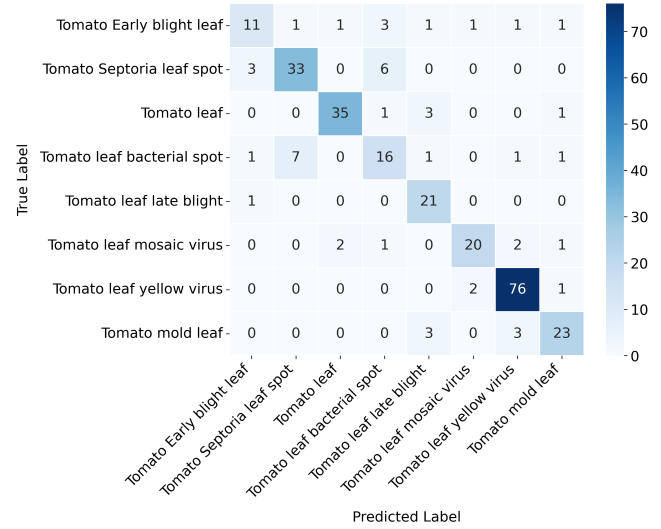
E. MODEL SIZE AND FLOPS

Table 5 summarizes model size and computation cost. MobileNetV3-Small-MoE increases the parameter count from 1.5260M to 3.4845M because all expert heads and the gating module must be stored. In contrast, FLOPs increase from 0.056G to 0.063G. This difference between parameter growth and FLOPs growth is consistent with the sparse top- k formulation in Eq. (18).

The observation is a clear trade-off. The proposed model is not smaller than the baseline in memory footprint. However, its active computational cost remains close to the baseline because only two out of four experts are activated per image. The implication is that the model is suitable when the main constraint is FLOPs or frame-level throughput, while



(a) MobileNetV3-Small-MoE.



(b) MobileNetV3-Small.

FIGURE 2. Confusion matrices of MobileNetV3-Small-MoE and MobileNetV3-Small. The matrices provide class-level diagnostic information, while Table 2 gives the cross-run performance summary.

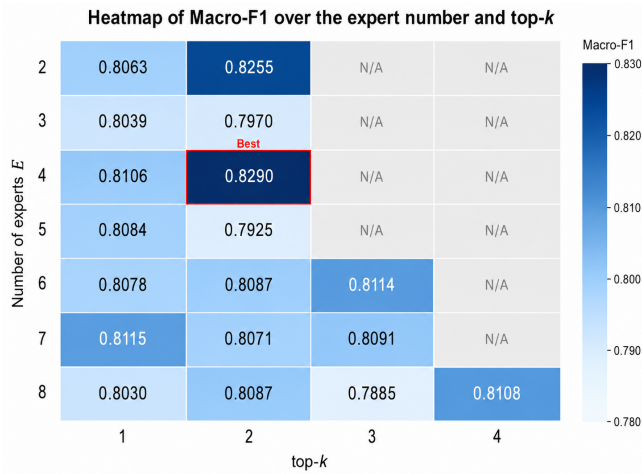


FIGURE 3. Heatmap of Macro-F1 over the number of experts E and the top- k routing width.

TABLE 5. Model size and FLOPs comparison.

Model	Params (M)	FLOPs (G)
MobileNetV3-Small	1.5260	0.0555
MobileNetV3-Small-MoE	3.4845	0.0634

memory-constrained deployment would require additional compression, expert sharing, or quantization.

F. EXPERT UTILIZATION

Fig. 4 illustrates the empirical expert utilization across the test set for both Linear Gating and MLP Gating mechanisms. In both settings, the routing network distributes samples across all four experts, indicating that the gating mechanism does

not collapse to a single dominant expert. Preventing expert collapse is important because such behavior would diminish the benefits of sparse routing, effectively reducing the Mixture-of-Experts (MoE) classifier to a dense architecture and leaving some expert parameters underutilized.

The observation is that both routers successfully utilize the available expert pool, preserving the intended sparse classifier structure compared with a collapsed router. To confirm whether this balanced utilization translates into meaningful semantic specialization on unseen data, Fig. 5 illustrates the class-wise expert activation heatmaps evaluated exclusively on the held-out test set. For Linear Gating (Fig. 5a), a clear and distinct pathological specialization emerges: Expert 1 (E_1) strongly dominates fungal and bacterial conditions (such as Early blight, Septoria leaf spot, and bacterial spot), E_2 primarily focuses on healthy tomato leaves, while E_3 explicitly handles viral infections (mosaic virus and yellow virus). In contrast, MLP Gating (Fig. 5b) exhibits a more distributed yet highly coordinated routing behavior. Under MLP Gating, viral classes remain tightly clustered under E_2 , while the responsibilities for blights, spots, and healthy leaves are shared and partitioned among E_1 , E_3 , and E_4 based on overlapping visual features. These distinct activation patterns provide strong empirical evidence that the routing mechanisms not only prevent parameter underutilization but also effectively partition the feature space according to the semantic and pathological characteristics of the diseases.

G. PARAMETER-MATCHED CLASSIFIER-SIDE BASELINES

The comparison with MobileNetV3-Small verifies the benefit of replacing the original dense classifier head with a sparse MoE head. However, this comparison alone cannot separate the effect of conditional routing from the effect of increased

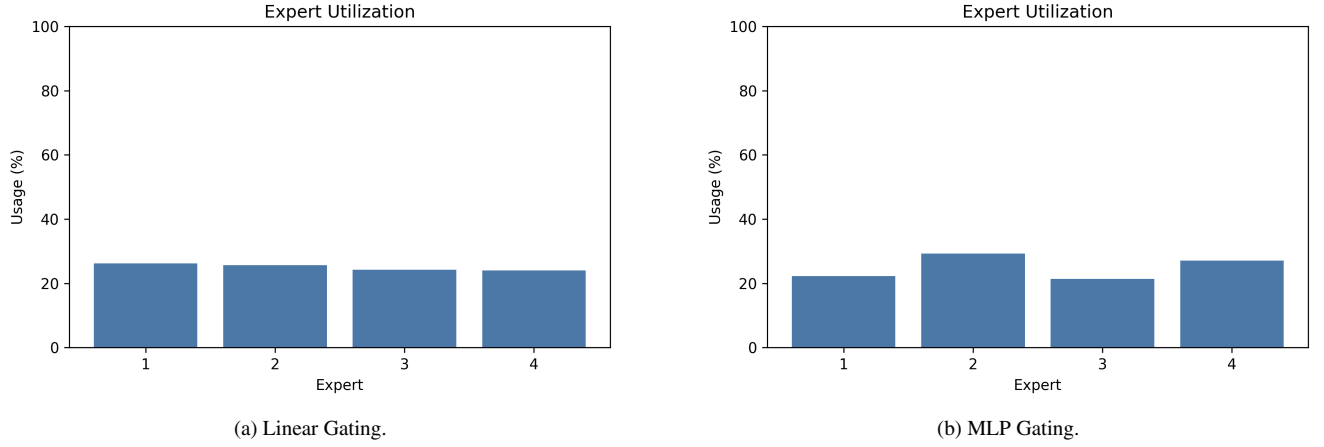


FIGURE 4. Expert usage under Linear Gating and MLP Gating. Both routers activate all experts during training. This result verifies utilization, but it should not be over-interpreted as proof of disease-category-specific expert behavior without class-wise routing analysis.

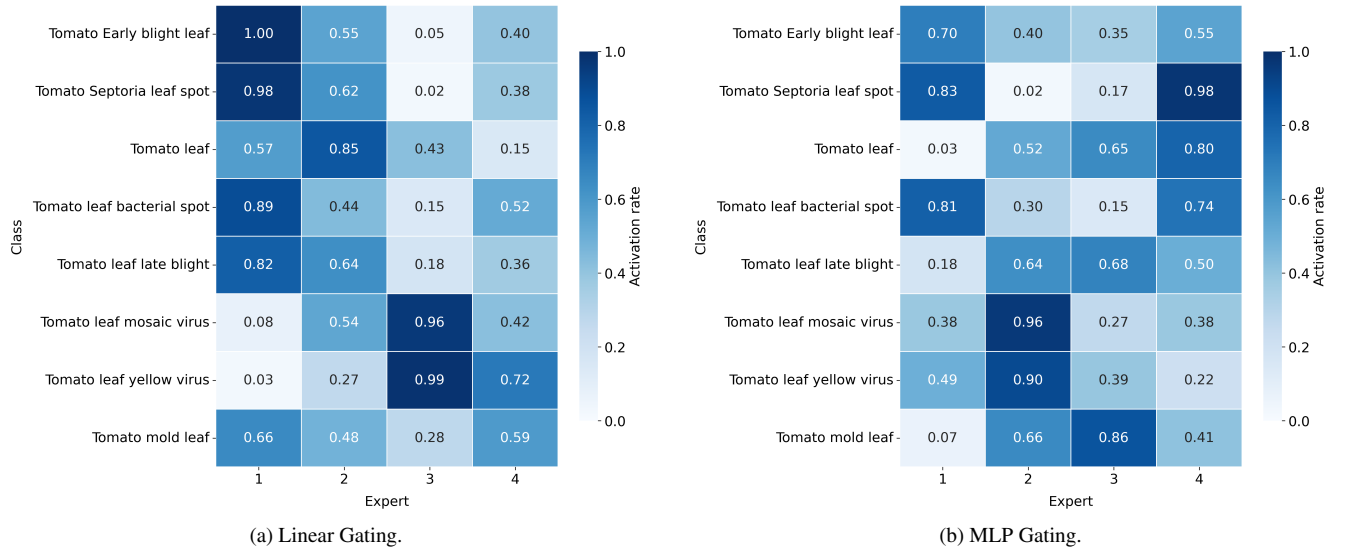


FIGURE 5. Class-wise expert routing heatmaps for Linear and MLP gating.

classifier-side parameter count. To address this issue, two parameter-matched classifier-side baselines are evaluated under the same training protocol.

The first baseline is MobileNetV3-Small with a widened MLP classifier head. Given the shared feature vector $\mathbf{u}_i$, the widened classifier is

$$\begin{aligned} \mathbf{r}_i &= \psi(\mathbf{W}_2^w \psi(\mathbf{W}_1^w \mathbf{u}_i + \mathbf{b}_1^w) + \mathbf{b}_2^w), \\ \hat{\mathbf{p}}_i^{\text{wide}} &= \text{softmax}(\mathbf{W}_c^w \mathbf{r}_i + \mathbf{b}_c^w). \end{aligned} \quad (23)$$

where the hidden width is selected such that the total parameter count is close to that of MobileNetV3-Small-MoE. Equation (23) tests whether the gain can be obtained by a larger dense classifier head without sparse routing.

The second baseline is a dense multi-branch classifier head without top- k selection. It uses the same expert branches as

the MoE model, but activates all branches and averages their outputs:

$$\begin{aligned} \mathbf{h}_i^{\text{dense}} &= \frac{1}{E} \sum_{e=1}^E q_e(\mathbf{u}_i), \\ \hat{\mathbf{p}}_i^{\text{dense}} &= \text{softmax}(\mathbf{W}_o^d \mathbf{h}_i^{\text{dense}} + \mathbf{b}_o^d). \end{aligned} \quad (24)$$

Equation (24) keeps the multi-branch capacity but removes input-dependent sparse routing; this baseline directly tests whether conditional expert activation is necessary.

MobileNetV3-Small-MoE outperforms the widened MLP head and the dense multi-branch head under comparable classifier-side parameter counts. This comparison indicates that the gain is not only caused by increasing the classifier width; input-dependent sparse routing also contributes to the measured Macro-F1 improvement.

TABLE 6. Parameter-matched classifier-side baseline comparison.

Model	Routing	Params (M)	FLOPs (G)	Macro-F1
MobileNetV3-Small	No	1.5260	0.0555	0.7799 ± 0.0276
Wide MLP head	No	3.1619	0.0572	0.7789 ± 0.0174
Dense multi-branch head	No	3.3314	0.0633	0.8145 ± 0.0183
MobileNetV3-Small-MoE	Top- <i>k</i>	3.4845	0.0634	0.8275 ± 0.0153

H. EDGE DEPLOYMENT ON RASPBERRY PI 5

To evaluate the deployability of the models on edge devices, we conducted experiments on a Raspberry Pi 5. All models were converted to the ONNX format, and inference was performed with a batch size of 1. For each model, we first applied 10 warm-up runs to stabilize execution, followed by 100 inference runs for measurement. The final reported latency is the average over these 100 runs. The results are presented in Table 7.

The experimental results indicate that the proposed MobileNetV3-Small-MoE model exhibits the highest memory footprint among all evaluated architectures. However, the overall memory consumption across all models remains low, rendering them compatible with the limited memory capacity of edge computing devices. The proposed MobileNetV3-Small-MoE achieves an inference latency of 8.9372 ms/image, compared with 6.2255 ms/image for the MobileNetV3-Small baseline. This result shows the cost of sparse classifier-side routing on the tested Raspberry Pi 5 platform: latency and memory increase, but the measured latency remains below 10 ms/image under the reported ONNX runtime setting.

I. COMPARISON WITH LIGHTWEIGHT BACKBONES

To verify whether the proposed design is competitive within the lightweight image-classification family, additional compact backbones are evaluated under the same data split, training schedule, input resolution, and augmentation policy. The comparison includes MobileNetV2, EfficientNet-B0 or EfficientNet-Lite, ShuffleNetV2, GhostNet, and SqueezeNet. These models are not used to redefine the proposed method; they provide an external reference for the accuracy-complexity operating point.

The comparison shows that MobileNetV3-Small-MoE improves over the MobileNetV3-Small dense baseline while keeping FLOPs close to the MobileNetV3-Small operating point. However, ShuffleNetV2 achieves higher Accuracy and Macro-F1, and GhostNet obtains a slightly higher Macro-F1. Therefore, the result should be interpreted as evidence for a classifier-side improvement over MobileNetV3-Small, not as universal superiority over all compact CNN backbones.

J. SEED-WISE STATISTICAL TEST

The reported mean and standard deviation are computed over five independent runs. However, mean ± standard deviation alone does not establish whether two models are statistically separated. If the same five data splits and random seeds

are used for two compared models, a paired test should be conducted on the seed-wise results.

Let $m_s^{(A)}$ and $m_s^{(B)}$ denote the metric value of two models under seed s , where $s = 1, \dots, S$ and $S = 5$. The paired difference is

$$\Delta_s = m_s^{(A)} - m_s^{(B)}. \quad (25)$$

Equation (25) removes the common variation caused by the same split or initialization condition, making the comparison more sensitive than an unpaired test.

For an approximate paired t -test, the test statistic is

$$t = \frac{\bar{\Delta}}{s_{\Delta}/\sqrt{S}}, \quad \bar{\Delta} = \frac{1}{S} \sum_{s=1}^S \Delta_s, \quad s_{\Delta}^2 = \frac{1}{S-1} \sum_{s=1}^S (\Delta_s - \bar{\Delta})^2. \quad (26)$$

Equation (26) tests whether the mean seed-wise difference is separated from zero under an approximate normality assumption on the paired differences.

Because only five paired seeds are available, the non-parametric Wilcoxon signed-rank test is additionally reported alongside the paired t -test. Statistical comparisons are conducted separately for Accuracy and Macro-F1 using identical seeds and data splits across all compared models. With $S = 5$, the Wilcoxon test is conservative and has limited power; therefore, cases in which only the paired t -test is significant are reported as partially supported rather than fully significant. Because 16 pairwise tests are reported in Table 10, the uncorrected p -values are interpreted as exploratory supporting evidence rather than confirmatory family-wise statistical proof. No multiple-comparison correction is applied; therefore, isolated $p < 0.05$ results should be interpreted cautiously. Table 10 summarizes the resulting mean paired differences and corresponding p -values.

Table 10 presents seed-wise paired statistical comparisons using identical random seeds and data splits across all models. The results should be interpreted cautiously because only five paired runs are available and 16 pairwise tests are reported. Under an uncorrected 0.05 threshold, multiple testing increases the chance of observing at least one false positive. Therefore, the reported p -values are used as exploratory evidence to support the mean-performance trends, not as confirmatory family-wise statistical proof.

For several comparisons, the paired t -test gives $p < 0.05$, whereas the Wilcoxon signed-rank test gives $p > 0.05$. This means that the mean paired difference is statistically separated under the parametric test, but the rank-based non-parametric evidence is not strong enough to confirm the same conclusion. Therefore, these cases are treated as partially

TABLE 7. Edge inference memory and latency on Raspberry Pi.

Model	Peak Memory (MB)	Latency (ms/img)
MobileNetV3-Small	81.8906	6.2255
Widened MLP head	88.5000	7.1110
Dense multi-branch head	90.1250	7.4633
MobileNetV3-Small-MoE	106.4375	8.9372

TABLE 8. Comparison with lightweight image-classification backbones under the same training protocol on the Plantdoc dataset.

Model	Classifier head	Params (M)	FLOPs (G)	Accuracy	Macro-F1
SqueezeNet	Dense	0.7626	0.2633	0.7670 $\pm$ 0.0208	0.7339 $\pm$ 0.0238
ShuffleNetV2	Dense	5.3614	0.5962	0.8737 $\pm$ 0.0102	0.8464 $\pm$ 0.0123
EfficientNet-B0/EfficientNet-Lite	Dense	4.0178	0.3846	0.8568 $\pm$ 0.0176	0.8281 $\pm$ 0.0254
MobileVit-Small	Dense	4.9428	1.4202	0.8674 $\pm$ 0.0103	0.8388 $\pm$ 0.0129
GhostNet	Dense	3.9118	0.1539	0.8596 $\pm$ 0.0161	0.8313 $\pm$ 0.0177
MobileNetV3-Small	Dense	1.5260	0.0555	0.8140 $\pm$ 0.0250	0.7799 $\pm$ 0.0276
MobileNetV3-Small-MoE	Sparse MoE	3.4845	0.0634	0.8596 $\pm$ 0.0124	0.8275 $\pm$ 0.0153

TABLE 9. Comparison with lightweight image-classification backbones under the same training protocol on the Slif-Tomato dataset.

Model	Classifier head	Params (M)	FLOPs (G)	Accuracy	Macro-F1
SqueezeNet	Dense	0.7626	0.2633	0.7670 $\pm$ 0.0208	0.7339 $\pm$ 0.0238
ShuffleNetV2	Dense	5.3614	0.5962	0.9715 $\pm$ 0.0056	0.9795 $\pm$ 0.0079
EfficientNet-B0/EfficientNet-Lite	Dense	4.0178	0.3846	0.8568 $\pm$ 0.0176	0.8281 $\pm$ 0.0254
MobileVit-Small	Dense	4.9428	1.4202	0.9835 $\pm$ 0.0033	0.9865 $\pm$ 0.0076
GhostNet	Dense	3.9118	0.1539	0.9850 $\pm$ 0.0109	0.9883 $\pm$ 0.0132
MobileNetV3-Small	Dense	1.5260	0.0555	0.9610 $\pm$ 0.0239	0.9562 $\pm$ 0.0287
MobileNetV3-Small-MoE	Sparse MoE	3.4845	0.0634	0.9850 $\pm$ 0.0095	0.9834 $\pm$ 0.0083

supported improvements rather than fully test-independent statistical gains.

Against MobileNetV3-Small, the Linear-Gated MoE shows positive mean differences in both Accuracy and Macro-F1. The paired t -test indicates a significant improvement in the mean, but the Wilcoxon test remains above the 0.05 threshold. A similar pattern appears for the widened MLP head and SqueezeNet comparisons. These results indicate a positive average gain relative to these baselines, while the statistical evidence is only partially significant. For the Dense Multi-Branch Head, MLP-Gated MoE, EfficientNet-B0, and GhostNet, both tests indicate that the performance differences are not statistically significant. Compared with ShuffleNetV2, the mean differences are negative, indicating that ShuffleNetV2 achieves higher average performance in this experiment, although the Wilcoxon test does not confirm a significant rank-based separation.

Overall, the statistical tests support a narrower conclusion: the proposed Linear-Gated MoE improves average performance over the MobileNetV3-Small baseline and some weaker classifier-side baselines, but this improvement is not consistently confirmed by both paired tests and should not be interpreted as family-wise statistical superiority. The main technical conclusion is therefore a controlled trade-off between accuracy and complexity at the MobileNetV3-Small operating point, rather than a general statistical superiority over all lightweight backbones.

V. DISCUSSION

The main result is that replacing the single MobileNetV3-Small classifier head with a sparse MoE head improves the average Accuracy and Macro-F1 on the tested tomato subset. The gain is stronger in Macro-F1 than in Accuracy, which is relevant for tomato disease classification because improvement on easy or frequent classes alone may not improve class-balanced disease recognition in field images. However, the paired statistical tests show mixed results for some comparisons: the paired t -test indicates a significant mean gain, whereas the Wilcoxon signed-rank test does not always yield a significant rank-based gain. In addition, the statistical table contains 16 pairwise tests, so uncorrected $p < 0.05$ results are vulnerable to multiple-comparison false positives. Therefore, the measured result should be interpreted as a positive average improvement with exploratory and partially supported statistical evidence, rather than as a uniformly significant advantage under all tests.

The comparison with the baseline also shows a concrete cost. The proposed model stores more than twice as many parameters, increasing from 1.526M to 3.484M. Therefore, it should not be described as uniformly lighter than MobileNetV3-Small. The correct claim is narrower: FLOPs increase moderately, from 0.0555G to 0.0634G, because only two out of four experts are activated per image. This distinction matters for deployment. If the edge device is limited by memory, the parameter increase is significant. If the device is primarily limited by multiply-accumulate operations or frame

TABLE 10. Seed-wise paired statistical tests across MoE variants and baseline models.

Comparison	Metric	Mean Δ	Paired t -test p	Wilcoxon p
Linear-Gated MoE vs MobileNetV3-Small	Accuracy	0.0456	0.0105	0.0625
	Macro-F1	0.0477	0.0057	0.0625
Linear-Gated MoE vs Widened MLP Head	Accuracy	0.0477	0.0258	0.0625
	Macro-F1	0.0487	0.0268	0.0625
Linear-Gated MoE vs Dense Multi-Branch Head	Accuracy	0.0098	0.2501	0.3125
	Macro-F1	0.0131	0.3046	0.3125
Linear-Gated MoE vs MLP-Gated MoE	Accuracy	0.0000	1.0000	1.0000
	Macro-F1	-0.0014	0.8429	1.0000
Linear-Gated MoE vs EfficientNetB0	Accuracy	0.0028	0.7893	1.0000
	Macro-F1	-0.0006	0.9713	1.0000
Linear-Gated MoE vs GhostNet	Accuracy	0.0000	1.0000	1.0000
	Macro-F1	-0.0038	0.6321	0.6250
Linear-Gated MoE vs ShuffleNetV2	Accuracy	-0.0140	0.0247	0.1250
	Macro-F1	-0.0189	0.0042	0.0625
Linear-Gated MoE vs SqueezeNet	Accuracy	0.0926	0.0023	0.0625
	Macro-F1	0.0937	0.0026	0.0625

throughput, the sparse routing design is preferable.

The gating comparison indicates that Linear Gating is sufficient for the current feature space. MLP Gating achieves the same mean accuracy and only a small increase in macro-F1. Since the standard deviations overlap, the present evidence does not justify a strong claim that MLP Gating is superior. A more defensible interpretation is that MobileNetV3-Small already produces features that can be routed by a linear decision surface, so the nonlinear gate does not provide a clearly separated benefit under the present training protocol.

The expert-number ablation gives a second design constraint. The best result appears at $E = 4, k = 2$, while larger expert pools do not consistently improve performance. This behavior is consistent with a finite-data classification problem: increasing the number of experts can reduce the effective sample size per expert and make the routing decision less stable. Therefore, the proposed model should be presented as a calibrated sparse classifier head, not as a scaling rule where more experts are always better.

The confusion matrices and expert-usage plots provide useful diagnostics, but they should be interpreted carefully. The confusion matrices help identify class-level failure modes in individual runs, while the mean and standard deviation tables provide stronger evidence for cross-run performance. The expert-usage plots show that all experts are activated, but they do not prove that each expert specializes in a disease category or a leaf-image pattern group. To support such a claim, future evaluation should add class-wise routing probability, expert-wise confusion matrices, or feature-space visualization conditioned on the selected expert.

Overall, MobileNetV3-Small-MoE is best positioned as a parameter-expanded classifier with low active computational cost for tomato leaf disease classification. Its technical value

lies in the controlled trade-off between classifier capacity and active computational cost. The current evidence supports three measured claims: sparse MoE improves mean classification performance over MobileNetV3-Small on the tested tomato subset; $E = 4, k = 2$ is the best tested routing configuration; and Linear Gating provides a simpler routing path with performance close to MLP Gating.

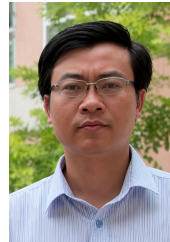
VI. CONCLUSION

This paper studied a sparse MoE classifier head on top of MobileNetV3-Small for tomato leaf disease classification. The proposed model keeps MobileNetV3-Small as a shared feature extractor and activates only the top- k expert heads for each input image. On the tested tomato subset derived from PlantDoc, MobileNetV3-Small-MoE improves accuracy and Macro-F1 compared with the MobileNetV3-Small baseline. The best tested configuration uses four experts and top-2 routing. Linear Gating achieves performance close to MLP Gating, making it the preferred routing structure when implementation simplicity is considered. The complexity analysis shows that the method increases stored parameters but keeps active computational cost close to the baseline. Future work should validate the model on additional plant disease datasets, add class-wise expert routing analysis, and measure latency and memory on target edge hardware.

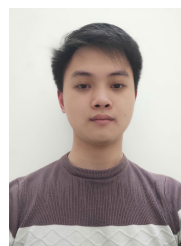
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